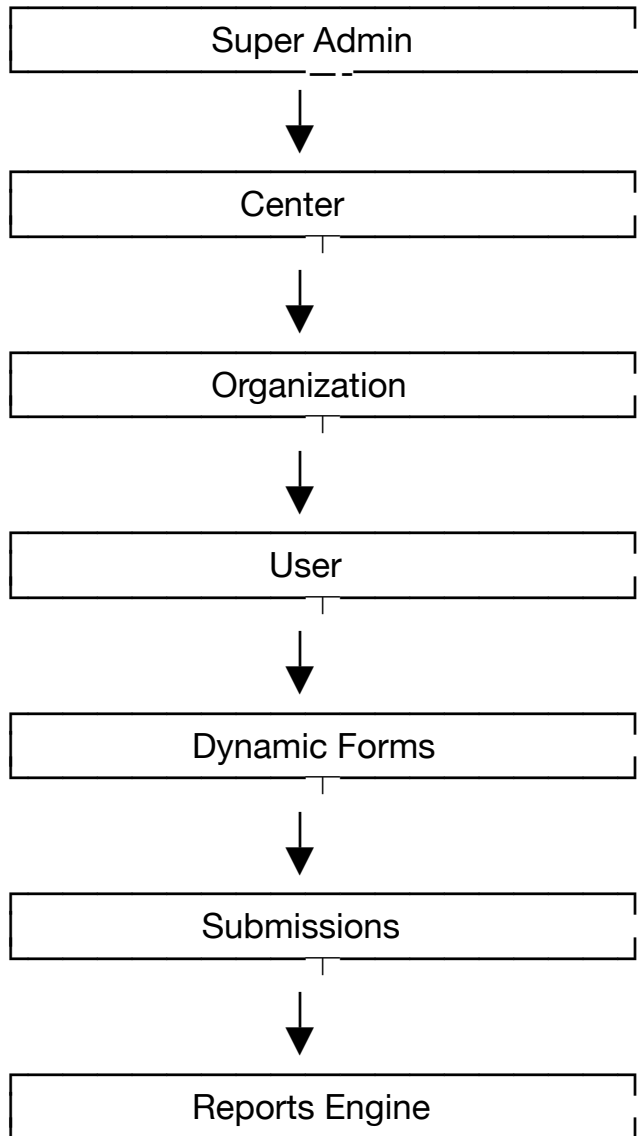


Project Architecture



Tech Stack

Frontend

React.js
Vite
Redux Toolkit
React Router

Axios
Tailwind
React Hook Form
Zod Validation

Backend

Node.js
Express.js
JWT Authentication
Passport.js
Socket.io (optional)

Database

MongoDB
Mongoose ODM
MongoDB is best because:

- Dynamic forms
- Dynamic fields
- Tree structures
- Flexible schema

Storage

Cloud
AWS S3
For:

- Images
- Attachments
- Documents

High Level Architecture

Frontend



API Gateway (Express)

- | — Auth Module
- | — User Module
- | — Center Module
- | — Organization Module
- | — Form Module
- | — Submission Module

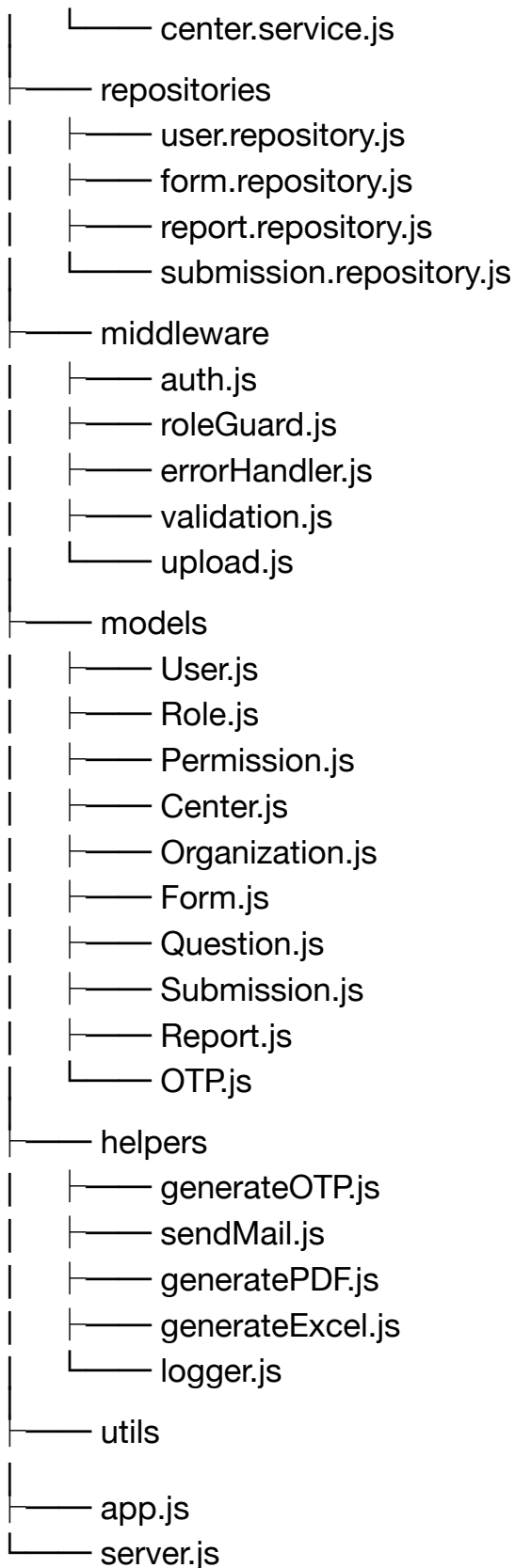
- Report Module
- OTP Module
- Notification Module



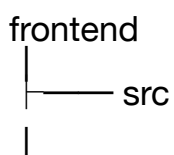
MongoDB

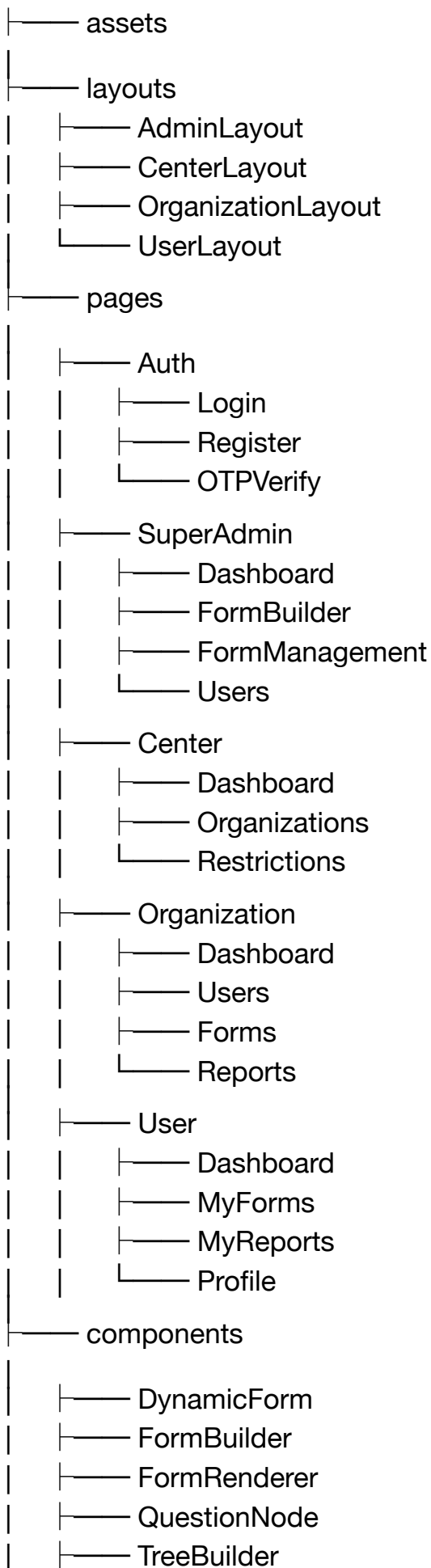
Backend Folder Structure

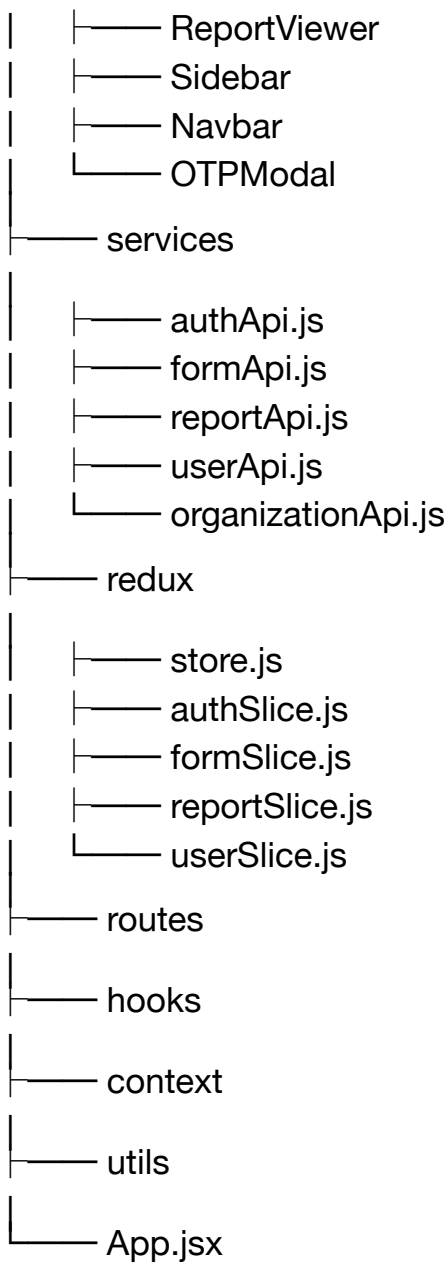
```
backend
├── src
├── config
│   ├── db.js
│   ├── jwt.js
│   ├── mail.js
│   └── cloudinary.js
├── routes
│   ├── auth.routes.js
│   ├── user.routes.js
│   ├── center.routes.js
│   ├── organization.routes.js
│   ├── form.routes.js
│   ├── report.routes.js
│   ├── submission.routes.js
│   └── otp.routes.js
├── controllers
│   ├── auth.controller.js
│   ├── form.controller.js
│   ├── report.controller.js
│   ├── submission.controller.js
│   ├── organization.controller.js
│   ├── center.controller.js
│   └── user.controller.js
├── services
│   ├── auth.service.js
│   ├── form.service.js
│   ├── report.service.js
│   ├── otp.service.js
│   └── organization.service.js
```



Frontend Folder Structure







Database Design

Users

```
{
  _id,
  name,
  email,
  phone,
  password,
  role,
  organizationId,
  centerId,
  isVerified,
  status
}
```

Centers

```
{
  _id,
  name,
  description,
  status
}
```

Organizations

```
{
  _id,
  centerId,
  name,
  status,
  restrictedForms:[]
}
```

Forms

```
{
  _id,
  title,
  description,
  visibility:"public/private",
  createdBy,
  version,
  status
}
```

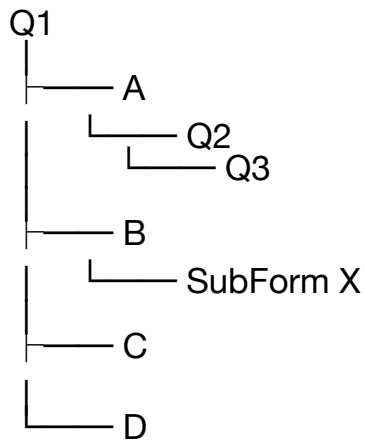
Questions

```
{
  _id,
  formId,
  question,
  type,
  validations,
  options
}
```

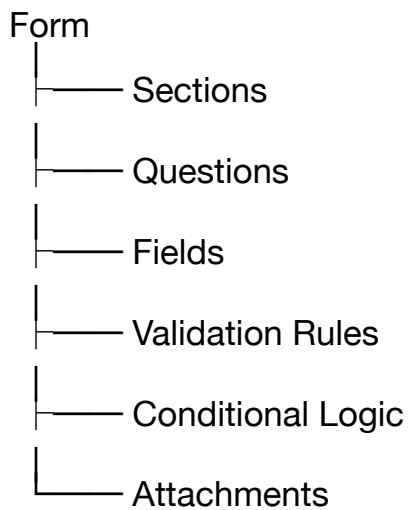
Conditional Tree Structure

```
{
  questionId:"Q1",
  option:"A",
  childQuestion:{
    questionId:"Q2"
  }
}
```

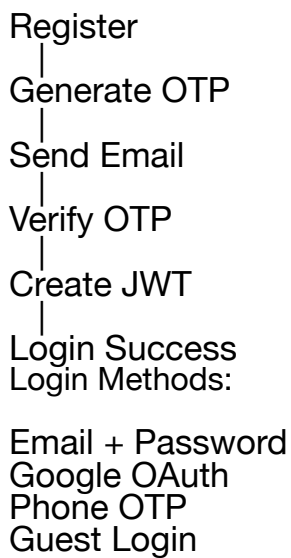
Example:



Form Builder Architecture



Authentication Flow



Public Form Distribution Flow

Super Admin
|
Create Form
|
Make Public
|
Center
|
Organizations
|
Users

Organization Custom Form Flow

Master Form
|
Clone Form
|
Organization Form
|
Custom Fields
|
Users
Never edit master form directly.

Use cloning/versioning.

Submission Flow

User
|
Fill Form
|
Validate
|
Submit
|
Store Responses
|
Generate Report
|
PDF/Excel/HTML

Report Engine

Generate:

Tabular Reports

Summary Reports

Analytics Reports

PDF Reports

Excel Reports

CSV Reports

HTML Reports

Filters:

Date

User

Organization

Center

Form

Status

Permissions (RBAC)

SUPER_ADMIN

- Create Forms
- Publish Forms
- Manage Centers
- Manage Everything

CENTER

- Manage Organizations
- Restrict Organizations

ORGANIZATION

- Manage Users
- Customize Forms

USER

- Fill Forms

APIs Required

Auth

POST /auth/register
POST /auth/login
POST /auth/google
POST /auth/send-otp
POST /auth/verify-otp

Forms

GET /forms
POST /forms
PUT /forms/:id
DELETE /forms/:id
POST /forms/publish
POST /forms/private

Organizations

POST /organizations
GET /organizations
PUT /organizations/:id
DELETE /organizations/:id

Reports

GET /reports
GET /reports/pdf
GET /reports/excel